Information Technologies for Industrial Engineers

เทคโนโลยีสารสนเทศสำหรับวิศวกรอุตสาหการ

Natural Language Processing (NLP)

What is NLP?

- A field of AI that makes human language intelligible to machines.
- NLP combines the power of linguistics and computer science to study the rules and structure of language.
- NLP creates intelligent systems capable of understanding, analyzing, and extracting meaning from text and speech.

NLP tasks (applications)

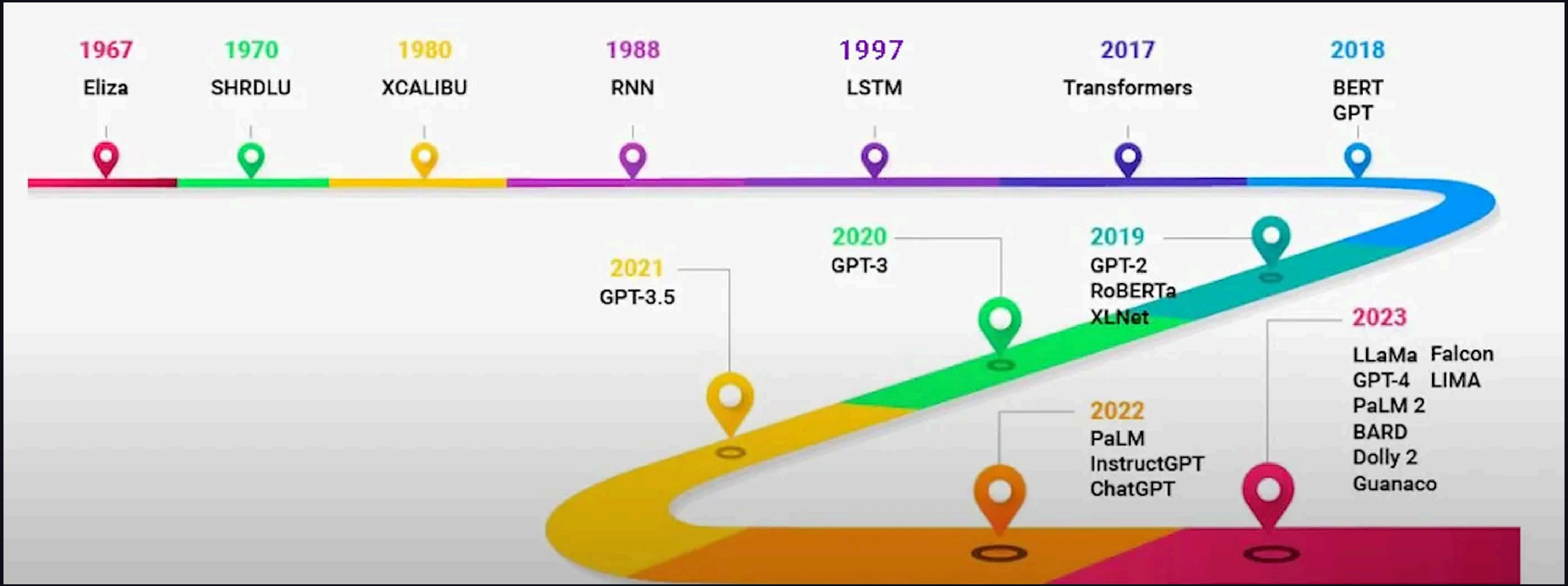
- Semantic Analysis
  - Classify text by polarity of opinion (positive, negative, neutral).
- Named Entity Recognition (NER)
  - Extract entities from within a text (names, places, organizations, email addresses, etc).
- Text Classification
  - Classify text into predefined categories (tags).
- Text Translation
- Text prediction (answering, chatbot)

Large Language Model

LLM

- Most advanced `tool` for NLP today,
- General-purpose language processing models
  - Pre-trained on extensive datasets covering a wide range of topics.
  - Understand the fundamental structures and semantics of human language.
- `Large`
  - Substantial amount training data
  - Billions or even trillions of parameters

History of NLP Technology/Tools



Chatbots and rule-based systems (1960s)

- `ELIZA`
  - The first chatbot ever built by humans.
- It can create an illusion of a conversation
  - by rephrasing user statements as questions using pattern matching and substitution methodology.
- Try it.

Recurrent Neural Networks (1980s)

- Neural networks that can "remember" previous input.
  - Using feedback loop.
- Suitable for natural language processing (NLP) tasks.
- Still, they are not good at retaining memory and suffer from long term memory loss.
  - Vanishing gradient.

Long Short Term Memory (1990s)

- Specialised type of RNN.
- Can remember information over long sequences.
- LSTM architecture
  - `input`, `forget`, `output` gates
  - These gates determined how much information should be memorised, discarded, or output at each step.

Gated Recurrent Network (2010s)

- Designed to solve some of the same problems as LSTMs
  - but with a simple and more streamlined structure.
- GRUs architecture
  - `update` and `reset` gate
  - The reduced gating in GRUs made them more efficient in terms of computation.

Attention Mechanism (2014)

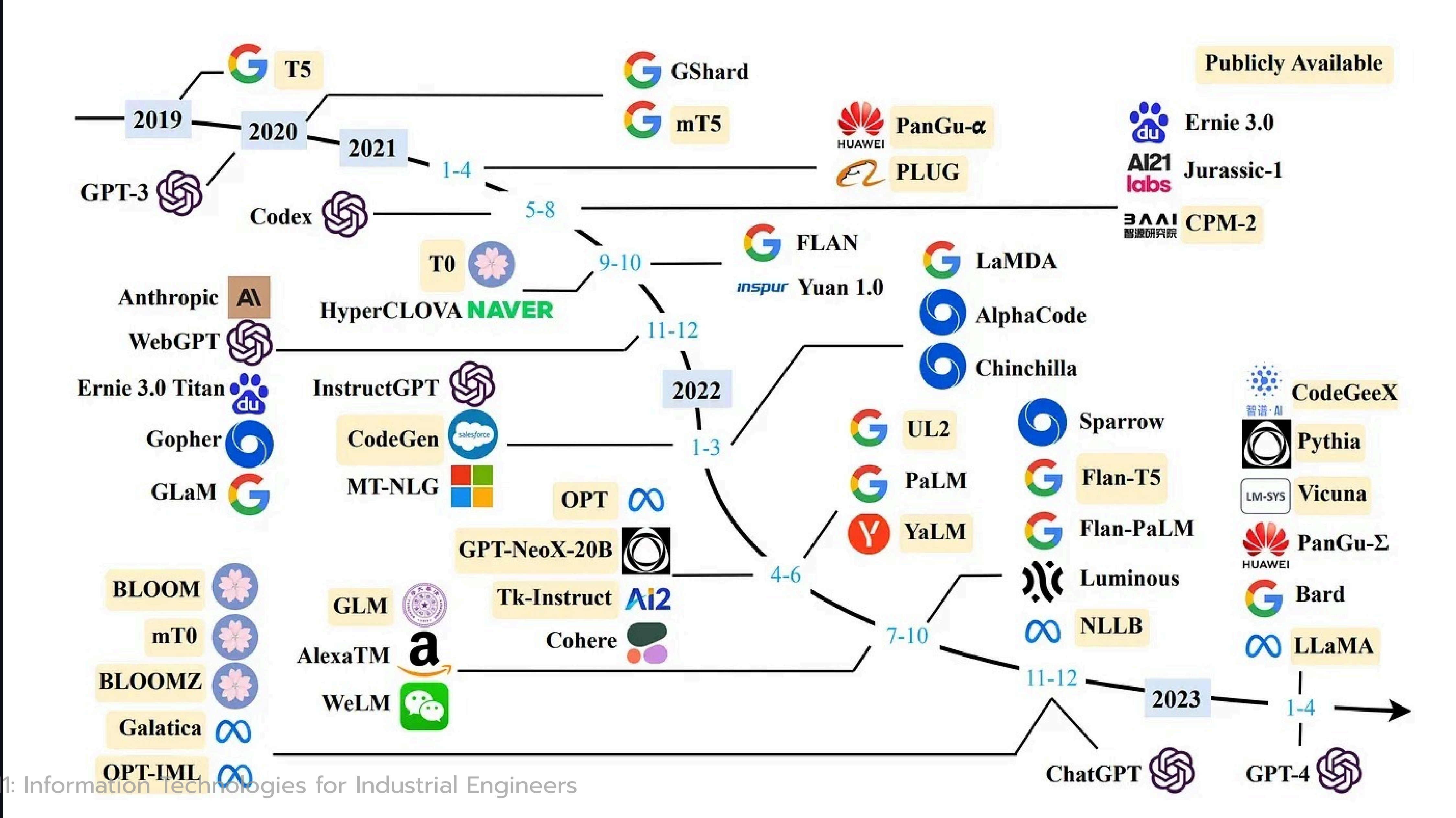
- RNN based variants LSTM and GRU are not great at retaining the context when it was far away.
- Attention allows the model to look back to the entire source sequence dynamically
  - Selecting different parts based on their relevance at each step of the output.
- Performs much better especially in longer sequences.

Transformers Architecture (2017)

- Architechture based on attention mechanism.
  - Stacked layers of (self) attention and feed-forward neural networks.
  - *Multi-head* attention
- Can focus on different parts of the input sentence simultaneously, capturing various contextual nuances.
- Can sequences in parallel rather than sequentially.
- Foundation for a new era of LLMs.
  - `GPT` ➡️ *Generative Pre-Trained Transformer*

Large Language Models (2018-onwards)

- Scaling...


Publicly Available
2019
2020
2021
2022
2023
1-4
5-8
9-10
11-12
1-3
4-6
7-10
11-12
1-4
T5
GPT-3
GShard
mT5
PanGu-α
HUAWEI
PLUG
Ernie 3.0
Jurassic-1
AI21 labs
CPM-2
Codex
T0
HyperCLOVA
NAVER
FLAN
inspur
Yuan 1.0
LaMDA
AlphaCode
Chinchilla
Anthropic
WebGPT
Ernie 3.0 Titan
Gopher
GLaM
InstructGPT
CodeGen
salesforce
MT-NLG
OPT
GPT-NeoX-20B
UL2
PaLM
YaLM
Sparrow
Flan-T5
Flan-PaLM
Luminous
NLLB
CodeGeeX
Pythia
LM-SYS
Vicuna
PanGu-Σ
HUAWEI
Bard
LLaMA
BLOOM
mT0
BLOOMZ
Galatica
OPT-IML
GLM
AlexaTM
WeLM
Tk-Instruct
Ai2
Cohere
ChatGPT
GPT-4

GPT-2

GPT-2

- Released in 2019.
- Milestone in AI's ability to generate coherent and contextually relevant text.
- Trained on 40GB of internet text data and consists of 1.5 billion parameters.
- Can perform various NLP tasks such as text completion, translation, and summarization without task-specific training.

More Info, Model Source

Setup

- `pnpm create vite@latest`
- ...
- `pnpm i @huggingface/transformers`
- `pnpm approve-builds`
    - Approve all packages.

Setup

- `./src/main.tsx`
  - Remove css import.
- `./index.html`
  - Add `PicoCSS`

Codes

- `./src/model.ts`
- `./src/App.tsx`

Problem

- UI hangup
- Reasons
  - Running heavy synchronous or async computations—like model loading and text generation—directly in the main thread
  - Blocking React's rendering and user interactions.

Solution - Web Worker

- A type of JavaScript that runs in the background on a separate thread.
- Does not block the main thread that handles UI and user interactions.
  - Prevent UI freezes.
- The main thread communicates with the worker using message passing.

Basic

- `index.html`
- `script.js`
- `worker.js`

Text-Generation Code

- `./src/types.tsx`
- `./src/App.tsx`
- `./src/worker.ts`

`SmolLM2-135M-Instruct`

- Part of the `SmolLM2` family
  - 135M parameters
  - Uses a transformer decoder architecture for text generation.
  - Trained on 2 trillion tokens from diverse datasets.
- Instruction-tuned for tasks like summarization, rewriting, question-answering, and conversational AI with prompt-based chat support.

Model

`Phi-3.5-mini-instruct-onnx-web`

- State-of-the-art open language model with ONNX support
  - Enabling high efficiency and deployment in web environments.
- Built using synthetic and filtered web datasets
- Optimized for instruction-following and generative tasks like Q&A, coding, math, and reasoning.

Model

Text-Translation

`nllb-200-distilled-600M`

- Multilingual machine translation model based on a distilled variant of Meta's `NLLB-200`,
- Translates single sentences between 200 languages.
- Utilizes a transformer architecture, distilled for efficiency and reduced resource usage compared to the original larger models.

Text-Translation Code

- `./src/types.tsx`
- `./src/App.tsx`
- `./src/worker.ts`

Google Gimini

> Let's build an email generator for students who want to write to their advisors.

You need this info.

```
API_KEY=
MODEL_ID=
```

Get `API_KEY`

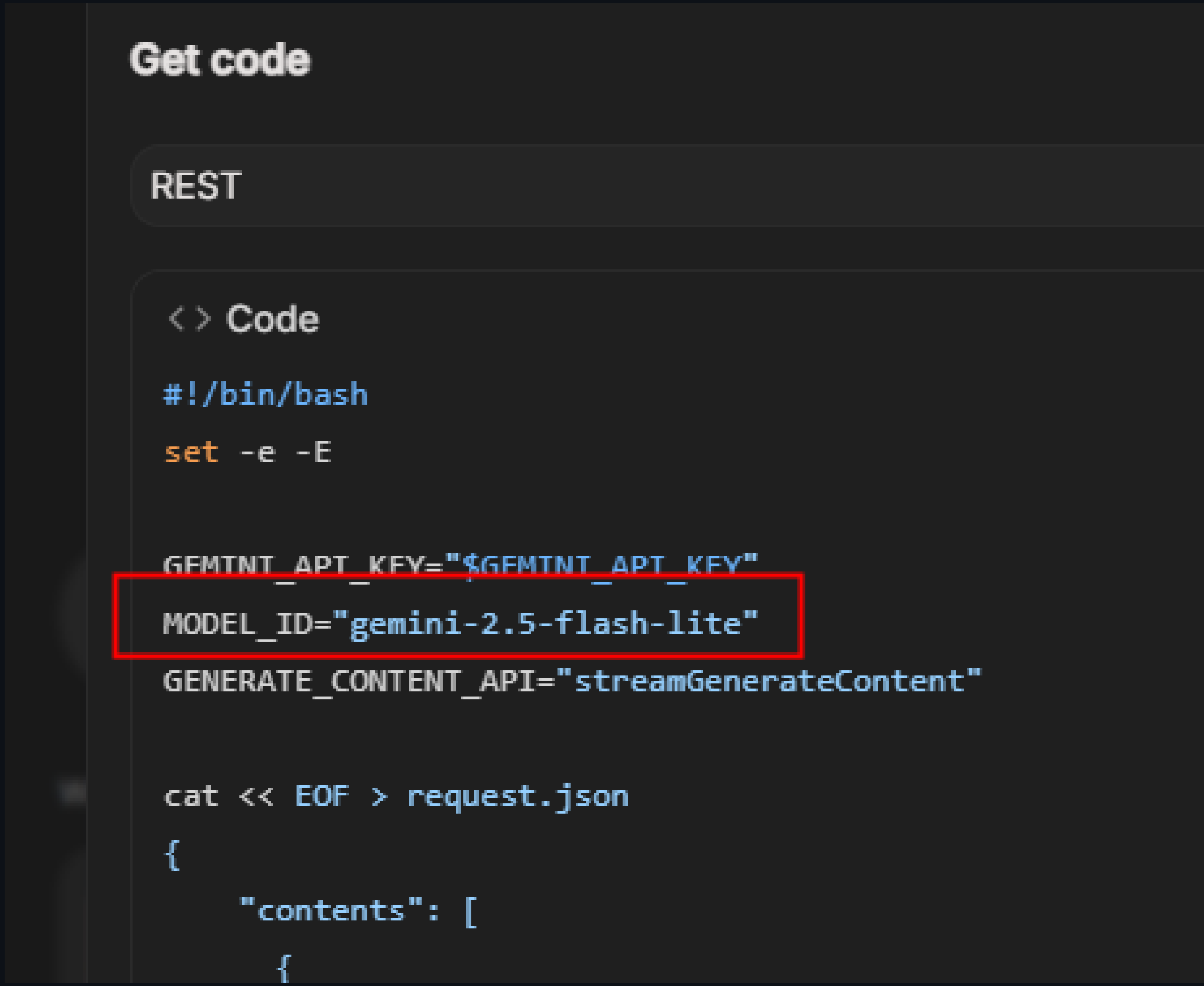
- Create a project in Google Cloud Platform
- Generate an API kep in Google AI Studio

Test Prompt: Model Settings

- Gemini Model Comparison
- `System Instructions`
    - The writer is a female student. The tone should be formal.
- `Prompt`
    - Write an email to the advisor in Thai. The topic is [ขอพบ วันจันทร์ นี้ ตอน 10 โมง ว่างมั้ย อยากปรึกษาเรื่องโปรเจค].
- Adjust `temperature` for more randomness.

Get `MODEL_ID`

- Click `Get Code`.
- Choose `REST`.
- Find the `MODEL_ID`.



Setup Vite Project

- `pnpm create vite@latest`
- ...

Setup

- `./.env`
  - Add `VITE_GEMINI_API_KEY=[API_KEY]` *(Change it)*
- `./src/main.tsx`
  - Remove css import.
- `./index.html`
  - Add `PicoCSS`

Codes

- `./src/model.ts`
  - Change your model on Line 2.
- `./src/App.tsx`